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$$20i + 21 = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = 21 \\ 2xy = 20 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 21 \\ xy = 10 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = 21 \\ x^2(-y)^2 = 100 \end{cases}$$

$$\lambda^2 - 21\lambda - 100 = 0$$

$$\Delta = (-21)^2 - 4 \cdot 100 = 841 = 29^2$$

$$\lambda_1 = \frac{21 + 29}{2} = 25$$

$$\lambda_2 = \frac{21 - 29}{2} = -4$$

$$x = \pm 5 \quad y = \pm 2$$

$$x + yi \in \{5 + 2i, -5 - 2i\}$$

References